Technology Stack

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, machine learning libraries, development tools, and deployment technologies used to build the Human Development Index (HDI) Prediction System. A well-chosen technology stack ensures that the application is scalable, maintainable, accurate, and easy to deploy.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	HTML5, CSS3, Bootstrap, Jinja2 Templates	Provides a responsive, user-friendly web interface for entering HDI indicators and displaying prediction results.

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	Python, Flask	Flask handles HTTP requests, processes user input, loads the trained machine learning model, and returns prediction results efficiently.
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	CSV Dataset, Pickle (.pkl) Model File	The CSV file stores the Human Development Index dataset for training, while the Pickle file stores the trained Linear Regression model for fast predictions.
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	Flask Local Server (or Render/Heroku if deployed)	Enables easy deployment and testing of the HDI Prediction web application. Flask provides lightweight and efficient hosting during development.
5	**Version Control & CI/CD** *(e.g., GitHub, GitLab, Jenkins)*	Git, GitHub	Supports version control, collaboration, project backup, and efficient source code management.

6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Pickle	NumPy and Pandas handle data processing, Scikit-learn builds and trains the Linear Regression model, Matplotlib and Seaborn generate visualizations, and Pickle saves the trained model for deployment.